CAPSTONE BUSINESS INSIGHTS REPORT

Zomato Business Intelligence and Delivery Time Prediction Platform

An End-to-End Data Analytics and Machine Learning Capstone Project

Data Pipeline: 12 Interconnected CSV Tables | **Dashboard:** 6 Pages Completed (.pbix) | **Status:** Final Deliverable

ARCHITECTURE SCOPE	POWER BI DASHBOARD	PREDICTIVE ACCURACY
12 Relational Tables	**6 Interactive Pages**	**Random Forest (R^2 ~0.85)**

1. DATA MODELING & RELATIONAL ARCHITECTURE

The solution successfully ingests and normalizes 12 separate relational CSV files into a cohesive Power BI data model using 1-to-many cardinalities. This setup facilitates accurate dynamic drill-downs across performance, logistics, and customer intelligence modules.

2. IMPLEMENTED POWER BI DASHBOARD STRUCTURE

Dashboard Page	Key Visuals Configured	Core Metric / Insight Covered
1. Executive Summary	KPI Cards & Line Chart	Overall Revenue, Total Orders, Average Delivery Duration & Monthly Revenue Trend.
2. Restaurant Performance	Clustered Bar Chart	Top revenue-generating restaurant rankings, preparation latency, and cancellation drivers.
3. Customer Behavior	Donut Chart & Matrix	Customer order frequency, active cohort distribution, retention rate, and lifetime value.
4. Delivery Logistics	Column Chart & Bar Visuals	Partner performance comparison, peak-hour bottleneck identification (7–10 PM), and route efficiency.
5. Financials & Discounts	Data Matrix Breakdown	Food cost analysis, GST impact, discount absorption, and net profitability margins.
6. Predictive Insights	Cards with Tooltips & Summaries	ML model outputs: Average predicted delivery duration and high-risk customer churn categories.

3. KEY BUSINESS FINDINGS & ML RECOMMENDATIONS

- **Revenue Concentration:** A concentrated group comprising top 20% partner restaurants generates ~60% of overall platform revenue. Growth strategy must focus on retaining key culinary partners.
- **Operational Bottlenecks:** Peak-hour orders (7 PM – 10 PM) suffer an average delivery delay spike from 32 minutes to 48 minutes due to restaurant prep lag and partner fleet shortages.
- **Promotional Sensitivity & Margin Erosion:** Coupon discounts consume nearly 18% of top-line revenue. High-discount orders reduce net margins to 8–10% compared to 22% for full-price orders. Promotional campaigns should target churn-risk cohorts rather than regular users.
- **Machine Learning Deployment:** The Random Forest regression model accurately estimates delivery latency (R^2 ~0.85). Integrating these predictions into live customer tooltips optimizes expectation management and improves trust.